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Single Point of Creation

Abstract

A system and method for generating non-fungible tokens (NFTs) associated with encrypted near-field communication (NFC) tags is described. In an example implementation, the system may include an NFT creation platform configured to receive a request to generate an NFT in association with an asset. A wallet having a private key is then generated. A smart contract associated with the NFT can then be configured. The smart contract may have a unique identifier, and the NFT creation platform may be further configured to encrypt the unique identifier of the smart contract using the private key of the wallet. The encrypted unique identifier of the smart contract may then be stored in a database.

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Background/Summary

BACKGROUND

[0001] The specification relates to electronic tracking systems. In particular, the specification relates to a system and method for generating non-fungible tokens (NFTs) associated with encrypted near-field communication (NFC) tags.

[0002] Electronic tracking systems enable manufacturers and consumers track products from creation to point of sale. Within these systems, users of electronic tracking systems may use NFC tags to uniquely identify a particular physical asset. Previously, physical asset inventory systems relied on barcodes to identify an asset. Additionally, non-fungible tokens (NFTs) are assets that have been tokenized via a blockchain. NFTs have been used to uniquely identify virtual assets, such as works of art, collectibles, trading cards, music, virtual worlds, and digital representations of physical assets. There lacks a mechanism to associate NFTs with encrypted NFC tags that may be generated by manufacturers of physical assets that wish to tokenize their physical assets.

SUMMARY

[0003] According to one innovative aspect of the subject matter described in this disclosure, a system for generating NFTs associated with encrypted NFC tags includes a processor and a memory storing instructions when executed cause the system to generate an NFT that has been associated with an encrypted NFC tag through a private key.

[0004] In general, a Web3 user mints NFTs that are digital representations of physical assets through a digital wallet application. First, a new digital wallet is created by the user. The digital wallet creates a private key only known to the user. Through the digital wallet, the user creates a smart contract for an NFT that has its own unique identifier. Each NFT that the Web3 user creates is uniquely identified and directly linked to one blockchain address. Each token has an owner, and the ownership information (i.e., the address in which the minted token resides) is publicly available. Ethereum is an example blockchain that records and mints NFTs for assets. A creator of a physical asset may tag their asset with another unique identifier that is linked to the minted NFT using the same private key combined with the unique identifier from the smart contract to encrypt the tag through near-field communication (NFC) using an NFC tag application. In this way, the NFC tag is forever linked with the NFT because the same unique identifier of the smart contract and the private key are used to encrypt the NFC tag.

[0005] As an example, a physical asset, such as a collectible sports jersey, may be embedded with an encrypted NFC tag using this process. The manufacturer may create an NFT by minting the token on the blockchain using a digital wallet that has a private key. Using the same private key to encrypt the NFC tag that stores the unique identifier of the smart contract associated with the NFT, the manufacturer has forever linked the NFT to the encrypted NFC tag. Upon sale or transfer of the collectible sports jersey, the point of sale device executes the smart contract and the ownership of the NFT is updated to the new owner after the sale is completed. Additionally, the new owner must have their own digital wallet associated with their own private key that encrypts the NFC tag embedded within the collectible sports jersey after the sale.

[0006] When NFT is created, the user who creates the NFT may name the asset. The name will be unique, and the naming can only be done once per asset identifier (ID). In case there are duplicate assets, the assets will be differentiated by the asset ID. Returning to the example above, the collectible sports jersey may be one of fifty that are made, each jersey having a unique asset ID.

[0007] Other aspects include corresponding methods, systems, apparatus, and computer program products for these and other innovative aspects.

[0008] The present disclosure may be particularly advantageous in a number of respects. First, the system enables new NFTs to be generated and physically associated with a physical asset through an encrypted NFC tag, enhancing the tracking of the physical asset. Second, the system implements a process to encrypt NFC tags using the private keys associated with minted NFTs for tagging physical assets, further enhancing management of the physical assets. Third, the system uses an

NFC tag application that works with a digital wallet application to manage the sale and/or transfer of NFTs and their associated encrypted NFC tagged physical assets, enabling real world commerce to occur at the same time an NFT is sold or transferred on the blockchain. This prevents forgeries and/or counterfeit assets to enter the marketplace. The foregoing advantages are illustrated by way of example, and it should be understood that the present disclosure may have numerous other advantages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings in which like reference numerals are used to refer to similar elements.

[0010] FIG. 1 is a high-level block diagram illustrating an example system for generating NFTs associated with encrypted NFC tags for users according to some implementations.

[0011] FIG. 2 is a high-level block diagram illustrating an example server according to some implementations.

[0012] FIG. 3 is a high-level block diagram illustrating an example digital wallet application and an example NFC tag application according to some implementations.

[0013] FIG. 4 is an example method of generating NFTs associated with encrypted NFC tags for users according to some implementations.

[0014] FIG. 5 is an example method of transferring NFTs associated with encrypted NFC tags for users according to some implementations.

DETAILED DESCRIPTION

[0015] FIG. 1 is a block diagram illustrating an example system **100** for generating NFTs associated with encrypted NFC tags for users according to some implementations. The illustrated system **100** includes a server **130** and a user device **120** that can be accessed by a user (e.g., a creator) **102**. While FIG. 1 illustrates one user device **120** and one server **130**, the present disclosure applies to a system architecture having one or more user devices **120** and/or one or more servers **130**. Furthermore, although FIG. 1 illustrates one network **124** coupled to the user devices **120** and the server **130**, in practice one or more networks **124** can be connected to these entities.

[0016] The network **124** can be a conventional type, wired or wireless, and may have numerous different configurations including a star configuration, token ring configuration or other configurations. Furthermore, the network **124** may include a local area network (LAN), a wide area network (WAN) (e.g., the Internet), and/or other interconnected data paths across which multiple devices may communicate. In some implementations, the network **124** may be a peer-to-peer network. The network **124** may also be coupled to or includes portions of a telecommunications network for sending data in a variety of different communication protocols. In some implementations, the network **124** includes Bluetooth communication networks or a cellular communications network for sending and receiving data including via short messaging service (SMS), multimedia messaging service (MMS), hypertext transfer protocol (HTTP), direct data connection, WAP, email, etc.

[0017] In some implementations, the user device **120** may be a computing device that includes a memory and a processor, for example a laptop computer, a desktop computer, a tablet computer, a mobile telephone, a personal digital assistant (PDA), a mobile email device, a portable game player, a game controller, a portable music player or other electronic device capable of accessing a network **124**. A user **102** can be a creator of the NFT and/or manufacturer of a physical asset **112** that is embedded with an encrypted NFC tag **108**. In the illustrated implementation, the user **102** interacts with the user device **120**, which is communicatively coupled to the network **124** via signal

line 126. The user device **120** includes a digital wallet application **132** and an NFC tag application **138**. The user **102** uses the digital wallet application **132** through the user device **120** to create a new digital wallet **118**, via signal line **122**, that is associated with a private key **104** unique to the digital wallet **118**.

[0018] The digital wallet application **132** can include code and routines for generating and managing a digital wallet for a user. In some implementations, the digital wallet application **132** can be implemented using hardware including a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). In some other implementations, the digital wallet application **132** can be implemented using a combination of hardware and software. In some implementations, the digital wallet application **132** may be stored in a combination of the devices and servers, or in one of the devices or servers. The digital wallet application **132** is described below in more detail with reference to FIGS. 2-5.

[0019] The NFC tag application **138** can include code and routines for configuring and managing an NFC tag for an asset. As illustrated in FIG. 1, a user device **120** may configure and encrypt a tag through near-field communication (NFC) via signal line **142** using an NFC tag application **138**. An asset **112** may include the encrypted NFC tag **108** that is configured to store an encrypted smart contract identifier (ID) **156**. The NFC tag application **138** requests the encrypted smart contract ID **156** from the server **130** based on the generated NFT **114**. The NFC tag application **138** then stores the encrypted smart contract ID **156** onto the encrypted NFC tag **108** using the NFT **114** via signal line **144**. In some implementations, the NFC tag application **138** can be implemented using hardware including a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). In some other implementations, the NFC tag application **138** can be implemented using a combination of hardware and software. In some implementations, the NFC tag application **138** may be stored in a combination of the devices and servers, or in one of the devices or servers. The NFC tag application **138** is described below in more detail with reference to FIGS. 2-5.

[0020] The server **130** can be a hardware server that includes a processor, a memory and network communication capabilities. The server **130** is communicatively coupled to the network **124** via signal line **154**. In some implementations, the server **130** sends and receives data to and from the user devices **120** via the network **124**. In some implementations, the server **130** is a e-commerce network server that provides an e-commerce platform service to users. For example, one or more users can buy and sell NFTs and the associated assets and publish the transactions on an e-commerce platform. In other implementations, the server **130** is a social network server that provides an e-commerce platform service to its users. A social network can be a type of social structure where the users may be connected by a common feature. The common feature includes relationships/connections, e.g., friendship, family, work, an interest, etc. The common features may be provided by one or more social networking systems including explicitly defined relationships and relationships implied by social connections with other online users, where the relationships form a social graph. In some examples, the social graph can reflect a mapping of these users and how they can be related. As an example, a friend of a user may purchase a purse for a user, where the purse is a physical asset embedded with an encrypted NFC tag that is associated with an NFT, a digital representation of the purse, that is uniquely identified on the blockchain. In some implementations, the server **130** is a game server that provides one or more online games to users. For example, a game may mint a pizza NFT that is associated with a redeemable gift card that is embedded with the encrypted NFC tag and redeemable for a pizza to be delivered to the user in real-life. In other implementations, the server **130** can be another type of server that provides other services to users, such as weapons tracking systems, jewelry inventory systems, art collection management systems, sports memorabilia tracking systems, and so on.

[0021] In the illustrated implementation, the server **130** includes an NFT creator application **134**, a digital wallet management module **136**, and a database **140**. The NFT creator application **134** provides functionality to create an NFT on a blockchain platform **150**, communicatively coupled to

the network **124** via signal line **148**. For example, the blockchain platform **150** may include a distributed ledger **152** on which the NFT may be minted by the NFT creator application **134**. After a digital wallet **118** is generated, a smart contract **110** is created, through the blockchain platform **150** via signal line **128**, that includes a smart contract identifier (ID) **106**. The non-fungible token **114** that is generated uses the same smart contract ID **106** via signal line **146**. The smart contract ID **106** is encrypted using the private key **104** associated with the digital wallet **118** of the user device **120** via signal line **116**. The encrypted smart contract ID **156** is stored in the database **140** and stored on the encrypted NFC tag **108** by the NFC tag application **138**.

[0022] The digital wallet management module **136** may include software and/or logic to provide the functionality for managing digital wallets generated by user devices **120**. For example, a digital wallet **118** may be generated by the digital wallet application **132** on the user device **120** and stored by the digital wallet management module **136** on the server **130** in the database **140**. As new digital wallets **118** are created, they may be managed and stored in database **140** by the digital wallet management module **136**. Additionally, as new NFTs are generated using the same digital wallet **118**, the unique identifiers of the NFTs (i.e., the unique addresses on the distributed ledger **152** of the blockchain platform **150**) are stored in association with the digital wallet **118** in the database **140**.

[0023] The database **140** can be a non-transitory memory that stores data for providing the functionality described herein. For example, the NFT creator application **134** receives the unique identifiers of the NFTs generated by the user **102** and stores the unique identifiers of the NFTs in association with the identifier of the digital wallet **118** associated with the user **102** in the database **140**. The database **140** may store other data associated with the server **130**. The database **140** may be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory or some other memory devices. In some implementations, the database **140** also includes a non-volatile memory or similar permanent storage device and media including a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, or some other mass storage device for storing information on a more permanent basis.

[0024] Referring to FIG. 2, the NFT creator application **134** and the digital wallet management module **136** are shown in more detail. FIG. 2 is a block diagram illustrating a server **130** that includes an NFT creator application **134**, digital wallet management module **136**, a processor **235**, a memory **237**, a communication unit **241**, an input device **243**, an output device **239**, a smart contract data store **140a**, a user data store **140b**, and a private key data store **140c** according to some examples. In FIG. 2 and the remaining figures, a letter after a reference number, e.g., “**140a**,” represents a reference to the element having that particular reference number. A reference number in the text without a following letter, e.g., “**140**,” represents a general reference to implementations of the element bearing that reference number. The components of the server **130** are communicatively coupled by a bus **220**. In some embodiments, the components of server **130** may reside in a third-party system connected to the server **130** over a network **124**.

[0025] The processor **235** includes an arithmetic logic unit, a microprocessor, a general purpose controller or some other processor array to perform computations and provide electronic display signals to a display device. Processor **235** processes data signals and may include various computing architectures including a complex instruction set computer (CISC) architecture, a reduced instruction set computer (RISC) architecture, or an architecture implementing a combination of instruction sets. Although FIG. 2 includes a single processor **235**, multiple processors **235** may be included. Other processors, operating systems, sensors, displays and physical configurations are possible.

[0026] The memory **237** stores instructions and/or data that may be executed by the processor **235**. The instructions and/or data may include code for performing the techniques described herein. The memory **237** may be a dynamic random access memory (DRAM) device, a static random access

memory (SRAM) device, flash memory or some other memory device. In some implementations, the memory **237** also includes a non-volatile memory or similar permanent storage device and media including a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, or some other mass storage device for storing information on a more permanent basis.

[0027] The communication unit **241** transmits and receives data to and from the server **130** or user device **120** depending upon where the digital wallet application **132** or NFC tag application **138** is stored. In some implementations, the communication unit **241** includes a port for direct physical connection to the network **124** or to another communication channel. For example, the communication unit **241** includes a USB, SD, CAT-5 or similar port for wired communication with the server **130** or user device **120**. In some implementations, the communication unit **241** includes a wireless transceiver for exchanging data with the server **130** or other communication channels using one or more wireless communication methods, including IEEE 802.11, IEEE 802.16, BLUETOOTH® or another suitable wireless communication method.

[0028] In some implementations, the communication unit **241** includes a cellular communications transceiver for sending and receiving data over a cellular communications network including via short messaging service (SMS), multimedia messaging service (MMS), hypertext transfer protocol (HTTP), direct data connection, WAP, e-mail or another suitable type of electronic communication. In some implementations, the communication unit **241** includes a wired port and a wireless transceiver. The communication unit **241** also provides other conventional connections to the network **124** for distribution of files and/or media objects using standard network protocols including TCP/IP, HTTP, HTTPS and SMTP, etc.

[0029] The input device **243** can be any device to input data to the server **130**. For example, the input device **243** is one of a game controller, a touch-sensitive device that receives gestural commands (e.g., a tap, double tap, scroll, swipe, etc.) from a user, a mouse, a track ball or other type of pointing device to input data into the user device **115**. In some implementations, the input device **243** includes one or more of a keyboard (e.g., a QWERTY keyboard), a microphone, a web camera or similar audio or video capture device.

[0030] The output device **239** can be a conventional type such as a liquid crystal display (LCD) or any other similarly equipped display device, screen or monitor. The output device **239** represents any device equipped to display electronic images and data as described herein. In some implementations, the input device **243** and the output device **239** are integrated to form a touch screen.

[0031] The data stores **140** may include a storage device that can be a non-transitory memory that stores data for providing the functionality described herein. The data stores **140** may be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory or some other memory devices. In some implementations, the storage device also includes a non-volatile memory or similar permanent storage device and media including a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, or some other mass storage device for storing information on a more permanent basis.

[0032] In some implementations, the data stores **140** stores inputs (e.g., keystrokes) provided by a user operating on the user device **120**. A smart contract data store **140a** may include metadata regarding smart contracts that are created by the user through the NFT creator application **134**, such as through the smart contract generation module **208**. A user data store **140b** may include metadata regarding users that have created NFTs through the NFT creator application **134**, such as through the NFT identification module **206**, as well as users that have been part of transactions that involve the sale or transfer of NFTs, such as through the NFT transactions processing module **202** and/or user data module **212** of the digital wallet management module **136**. Once an NFT is generated, its address on the blockchain may be determined using the NFT identification module **206** and then

the associated smart contract identifier may be stored in the smart contract data store **140a**, for example. A private key data store **140c** may include metadata regarding private keys that are generated through the creation of digital wallets, such as through the digital wallet management module **136**, including the control module **214** and/or user data module **212**. One or more databases **140** may store any other data for providing the functionality described herein. The databases **140** may include a blockchain that stores transactional data in a digital ledger. These transactions may be performed and/or processed by the NFT transactions processing module **202**, for example. Messages to the blockchain platform, such as blockchain platform **150** illustrated in FIG. **1**, may be generated through the blockchain communications module **204**, for example. Such messages may include a request to mint an NFT, a request to transfer ownership of an NFT, and so forth.

[0033] In the illustrated implementation shown in FIG. **2**, the digital wallet management module **136** includes a control module **214** and a user data module **212**. These components of the digital wallet management module **136** are communicatively coupled to each other via the bus **220**.

[0034] The control module **202** can be software including routines for handling communications between the digital wallet management module **136** and other components of the server **130**. In some implementations, the control module **202** can be a set of instructions executable by the processor **235** to provide the functionality described below for handling communications between the digital wallet management module **136** and other components of the server **130**. In some other implementations, the control module **202** can be stored in the memory **237** of the server **130** and can be accessible and executable by the processor **235**. The control module **202** may be adapted for cooperation and communication with the processor **235** and other components of the server **130** via the bus **220**.

[0035] The control module **202** sends and receives data, via the communication unit **241**, to and from other entities of the system **100**. For example, the control module **202** receives, via the communication unit **241**, one or more packets from other user devices **120** operated by other users performing transactions with their digital wallets. In some implementations, the control module **202** handles communications between components of the digital wallet management module **136**. For example, the control module **202** receives data describing a purchase of an NFT and its associated real-world asset and sends the data to the user data module **212** for presenting the purchase of the NFT to the user. The purchase may incorporate the digital representation of the real-world asset into the inventory of the user's digital wallet through the digital wallet management module **136**, in an embodiment.

[0036] In some implementations, the control module **214** receives data from components of the digital wallet management module **136** and stores the data in the database **140**. For example, the control module **202** receives a transaction result and stores the transaction result in the database **140**.

[0037] The user data module **212** can be software including routines for generating data associated with users, such as interactions in a software service and/or NFT & asset purchases and/or transfers. In some implementations, the user data module **212** can be a set of instructions executable by the processor **235** to provide the functionality described below for generating data for tracking data generated by users. In some other implementations, the user data module **212** can be stored in the memory **237** of the server **130** and can be accessible and executable by the processor **235**. The user data module **212** may be adapted for cooperation and communication with the processor **235** and other components of the server **130** via the bus **220**.

[0038] The foregoing description of FIG. **2** illustrates an example scenario where two users (e.g., the first user and the second user) perform a transaction on an NFT in the same blockchain platform. It should be understood that in some implementations more than two users (e.g., an owner sells an NFT and the associated asset to more than one buyer) can participate in a single transaction simultaneously and similar operations can be performed. In an embodiment, fractional ownership of an NFT and its associated asset may be configured in the smart contract for the NFT.

Additionally, multiple NFTs may be combined to form a new NFT by a user and the new smart contract that is generated may be stored in the smart contract data store **140a** to be made available for purchase. For example, a set of collectible toys may be combined as one NFT such that each individual NFT for each collectible toy is merged into the NFT for the set.

[0039] FIG. 3 is a block diagram illustrating an example digital wallet application and an example NFC tag application. As shown in FIG. 3, a digital wallet application **132** may include a wallet selection engine **302**, a smart contract data module **304**, an NFT creation module **306**, an NFT management module **308**, a user input module **310**, a private key generation module **312**, a user interface engine **314**, and an encryption module **316**, in an embodiment. An NFC tag application **138** may include an NFC tag data management module **318**, in an embodiment.

[0040] The wallet selection engine **302** may include software and/or logic to provide the functionality for enabling a user to select a digital wallet associated with a particular blockchain platform. For example, a digital wallet for Ethereum may be completely separate from a digital wallet for Bitcoin. NFTs minted on the Ethereum blockchain may have different properties and/or metadata from NFTs minted on the Bitcoin blockchain. The wallet selection engine **302** thus enables the user to decide which blockchain platform to use.

[0041] The smart contract data module **304** may include software and/or logic to provide the functionality for handling and managing smart contract data that is generated from the creation of NFTs. For example, smart contract data may include various stipulations and/or contractual clauses that are modifiable by the user. As an example, fractional ownership of an NFT may be limited to a certain number of fractional shares as outlined in the smart contract for the NFT. The smart contract data module **306** enables the creator of the NFT to generate the specific smart contract terms they wish to associate with the NFT and have them stored in the data store **140**, in an embodiment.

[0042] The NFT creation module **306** may include software and/or logic to provide the functionality for managing the creation of the NFT on the blockchain. For example, the process of minting an NFT on the blockchain may vary, with different fees associated based on the blockchain platform and marketplaces. Ethereum, the most popular blockchain for NFTs, has higher gas fees, or transaction fees to mint an NFT on a blockchain, than less popular blockchains like Solana and Cardano. Additionally, some marketplaces charge a fee to list an NFT for sale. Other marketplaces charge a flat listing fee as a percentage of sales. Further, some marketplaces charge account setup fees. Some marketplaces offer an option for “lazy minting,” meaning that no gas fees are charged until an NFT that is listed is purchased, and at that point, the buyer pays the gas fees to have the NFT minted. Lastly, other marketplaces have “gasless” minting where the token you mint will be immediately found on the Ethereum blockchain. Gasless transactions are possible due to meta transactions that allow different users to do transactions on the public blockchain with zero transaction fees. In this framework, a third party relayer, who pays the gas and completes the transaction, uses proxy contracts to perform regular blockchain transactions. Thus, the different ways to get a new NFT to be minted may be managed by the NFT creation module **306**.

[0043] The NFT management module **308** may include software and/or logic to provide the functionality for managing transactions associated with the NFT. For example, the owner may enable fractional shares of the NFT to be bought and sold or otherwise transferred. As another example, NFTs may also be promoted on various marketplaces, and thus management of the promotional campaigns may be performed through the NFT management module **308**.

[0044] The user input module **310** may include software and/or logic to provide the functionality for receiving user input at the digital wallet application **132**. For example, as the options on when and how to mint NFTs, options on promotion of NFTs, and other choices are requested from a creator user, the user input module **310** provides user interfaces to capture the user input. The user interface engine **314** may include software and/or logic to provide the functionality for modifying user interfaces and presenting them on user devices **120**, in an embodiment.

[0045] The private key generation module **312** may include software and/or logic to provide the

functionality for generating a private key for each digital wallet. Various cryptographic methods and/or techniques may be used to generate private keys.

[0046] The encryption module **316** may include software and/or logic to provide the functionality for encrypting data using a private key and a public key. For example, data encrypted with the public key can only be decrypted with the private key. Thus, the encryption module **316** may encrypt the smart contract ID so that only the creator, using the private key generated by the private key generation module **312**, can decrypt the encrypted smart contract ID.

[0047] The NFC tag data management module **318** may include software and/or logic to provide the functionality for managing the data stored on the NFC tag. For example, the creator user, after creating a digital wallet, generating a private key, and generating an NFT on a blockchain, may use the encryption module **316** to encrypt the smart contract ID associated with the NFT and use the NFC tag application **138** to store the encrypted smart contract ID onto a tag embedded in an asset. The NFC tag data management module **318** enables the NFC tag application **138** to store the encrypted smart contract ID onto the tag. As ownership of the NFT and the associated asset is transferred to a new owner's digital wallet, the new owner's private key will be used to encrypt the smart contract ID such that the NFC tag application **138** modifies the encrypted NFC tag data through the NFC tag data management module **318**.

[0048] FIG. **4** is an example method of generating NFTs associated with encrypted NFC tags for users according to some implementations. In method **400**, step **402** includes a request being received to create a smart contract in association with a user-selected NFT on a blockchain platform where the request includes a private key.

[0049] At step **404**, the smart contract is generated on the blockchain platform, where the smart contract has a unique identifier associated with the user-selected NFT.

[0050] At step **406**, the smart contract is encrypted using the private key.

[0051] At step **408**, the encrypted smart contract unique identifier is caused to be stored on a tag through NFC, where the tag has a unique identifier.

[0052] At step **410**, an association is generated between the tag unique identifier and the NFT based on a unique identifier associated with the NFT.

[0053] At step **412**, the association is stored between the tag unique identifier and the NFT unique identifier in a database.

[0054] FIG. **5** is an example method of transferring NFTs associated with encrypted NFC tags for users according to some implementations. In method **500**, step **502** begins with encrypted data from a tag being received using NFC, where the tag has a unique identifier and the tag is embedded in an asset.

[0055] At step **504**, the tag data is decrypted using a private key.

[0056] At step **506**, an NFT associated with the decrypted tag data is identified on a blockchain platform, where the NFT is associated with the asset.

[0057] At step **508**, a smart contract associated with the NFT on the blockchain platform is retrieved, where the smart contract has a unique identifier associated with an NFT unique identifier.

[0058] At step **510**, a set of transaction options is provided based on the smart contract.

[0059] At step **512**, user input is received associated with the provided set of transaction options.

[0060] At step **514**, the smart contract is executed based on the received user input.

[0061] In the above description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the specification. It will be apparent, however, to one skilled in the art that the invention can be practiced without these specific details. In other instances, structures and devices are shown in block diagram form in order to avoid obscuring the description. For example, the present embodiment is described in one embodiment below primarily with reference to user interfaces and particular hardware. However, the present embodiment applies to any type of computing device that can receive data and commands, and any peripheral devices providing services.

[0062] Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the description. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0063] Some portions of the detailed descriptions that follow are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers or the like.

[0064] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the following discussion, it is appreciated that throughout the description, discussions utilizing terms including “processing” or “computing” or “calculating” or “determining” or “displaying” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0065] The present embodiment of the specification also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise a general-purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, including, but is not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, flash memories including USB keys with non-volatile memory or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0066] The specification can take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment containing both hardware and software elements. In a preferred embodiment, the specification is implemented in software, which includes but is not limited to firmware, resident software, microcode, etc.

[0067] Furthermore, the description can take the form of a computer program product accessible from a computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution system. For the purposes of this description, a computer-usable or computer readable medium can be any apparatus that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

[0068] A data processing system suitable for storing and/or executing program code will include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code in order to reduce the number of times code must be retrieved from bulk storage during execution.

[0069] Input/output or I/O devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the system either directly or through intervening I/O controllers.

[0070] Network adapters may also be coupled to the system to enable the data processing system to become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.

[0071] Finally, the algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear from the description below. In addition, the specification is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the specification as described herein.

[0072] The foregoing description of the embodiments of the specification has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the specification to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the disclosure be limited not by this detailed description, but rather by the claims of this application. As will be understood by those familiar with the art, the specification may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Likewise, the particular naming and division of the modules, routines, features, attributes, methodologies and other aspects are not mandatory or significant, and the mechanisms that implement the specification or its features may have different names, divisions and/or formats. Furthermore, as will be apparent to one of ordinary skill in the relevant art, the modules, routines, features, attributes, methodologies and other aspects of the disclosure can be implemented as software, hardware, firmware or any combination of the three. Also, wherever a component, an example of which is a module, of the specification is implemented as software, the component can be implemented as a standalone program, as part of a larger program, as a plurality of separate programs, as a statically or dynamically linked library, as a kernel loadable module, as a device driver, and/or in every and any other way known now or in the future to those of ordinary skill in the art of computer programming. Additionally, the disclosure is in no way limited to implementation in any specific programming language, or for any specific operating system or environment. Accordingly, the disclosure is intended to be illustrative, but not limiting, of the scope of the specification, which is set forth in the following claims.

Claims

1. A computer-implemented method, comprising: receiving a request to generate a non-fungible token (NFT) in association with an asset; generating a wallet having a private key; configuring a smart contract associated with the NFT, the smart contract having a unique identifier; encrypting the unique identifier of the smart contract using the private key of the wallet; and storing the encrypted unique identifier of the smart contract in a database.
2. The method of claim 1, further comprising: receiving a request to associate the NFT to the asset; retrieving the encrypted unique identifier of the smart contract from the database; and storing the encrypted unique identifier of the smart contract on a tag embedded within the asset.
3. The method of claim 2, wherein the encrypted unique identifier of the smart contract is stored on the tag embedded within the asset using near-field communication.
4. The method of claim 2, wherein the tag has a unique identifier, the method further comprising: generating an association between the unique identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract; and storing the association between the unique

identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract in the database.

5. The method of claim 3, further comprising: scanning, by a user device, the tag storing the encrypted unique identifier; and decrypting the encrypted unique identifier using the private key.

6. The method of claim 5, further comprising: identifying the NFT on a blockchain platform based on the decrypting; retrieving the smart contract from the database associated with the NFT; and providing a set of transaction options to the user device based on the smart contract.

7. The method of claim 6, further comprising: receiving user input associated with the provided set of transaction options at the user device; and executing the smart contract based on the received user input.

8. A system comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to the execution of the instructions by one or more processors, cause the one or more processors to perform operations including: receiving a request to generate a non-fungible token (NFT) in association with an asset; generating a wallet having a private key; configuring a smart contract associated with the NFT, the smart contract having a unique identifier; encrypting the unique identifier of the smart contract using the private key of the wallet; and storing the encrypted unique identifier of the smart contract in a database.

9. The system of claim 8, wherein the operations further comprise: receiving a request to associate the NFT to the asset; retrieving the encrypted unique identifier of the smart contract from the database; and storing the encrypted unique identifier of the smart contract on a tag embedded within the asset.

10. The system of claim 9, wherein the encrypted unique identifier of the smart contract is stored on the tag embedded within the asset using near-field communication.

11. The system of claim 9, wherein the tag has a unique identifier and wherein the operations further comprise: generating an association between the unique identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract; and storing the association between the unique identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract in the database.

12. The system of claim 10, wherein the operations further comprise: scanning, by a user device, the tag storing the encrypted unique identifier; and decrypting the encrypted unique identifier using the private key.

13. The system of claim 12, wherein the operations further comprise: identifying the NFT on a blockchain platform based on the decrypting; retrieving the smart contract from the database associated with the NFT; and providing a set of transaction options to the user device based on the smart contract.

14. The system of claim 13, wherein the operations further comprise: receiving user input associated with the provided set of transaction options at the user device; and executing the smart contract based on the received user input.

15. One or more non-transitory computer-readable storage media storing one or more sequences of instructions which, when executed using one or more processors, cause the one or more processors to execute a method comprising: receiving a request to generate a non-fungible token (NFT) in association with an asset; generating a wallet having a private key; configuring a smart contract associated with the NFT, the smart contract having a unique identifier; encrypting the unique identifier of the smart contract using the private key of the wallet; and storing the encrypted unique identifier of the smart contract in a database.

16. The one or more non-transitory computer-readable storage media of claim 15, wherein the method further comprises: receiving a request to associate the NFT to the asset; retrieving the encrypted unique identifier of the smart contract from the database; and storing the encrypted unique identifier of the smart contract on a tag embedded within the asset.

17. The one or more non-transitory computer-readable storage media of claim 16, wherein the encrypted unique identifier of the smart contract is stored on the tag embedded within the asset using near-field communication.

18. The one or more non-transitory computer-readable storage media of claim 16, wherein the tag has a unique identifier, the method further comprising: generating an association between the unique identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract; and storing the association between the unique identifier of the tag with the NFT based on the encrypted unique identifier of the smart contract in the database.

19. The one or more non-transitory computer-readable storage media of claim 17, the method further comprising: scanning, by a user device, the tag storing the encrypted unique identifier; decrypting the encrypted unique identifier using the private key. identifying the NFT on a blockchain platform based on the decrypting; retrieving the smart contract from the database associated with the NFT; and providing a set of transaction options to the user device based on the smart contract.

20. The one or more non-transitory computer-readable storage media of claim 19, the method further comprising: receiving user input associated with the provided set of transaction options at the user device; and executing the smart contract based on the received user input.
